

RESEARCH ARTICLE

Zero-Shot Visual Anomaly Detection on a Quadruped Robot Using State of the Art Visual Language Models

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ABSTRACT Anomaly detection in dynamic real-world environments remains a significant challenge for robotic systems, largely because traditional vision and rule-based methods struggle to interpret complex semantic contexts without task-specific training. Recent advances in Visual Language Models (VLMs) offer new opportunities for robots to perform zero-shot, context aware perception; however, their practical deployment on mobile robotic platforms remains underexplored. In this study, a quadruped robot autonomously patrolled both indoor and outdoor environments following a predefined trajectory and schedule to perform real-time, zero-shot anomaly detection. The robot was equipped with an onboard RGB camera and integrated with a VLM to interpret visual data without any fine tuning. Anomalies were identified directly through prompt-based semantic reasoning, enabling detection of misplaced objects, structural defects, and environmental hazards. The overall framework was implemented under ROS2 to ensure seamless communication, control, and real-time decision making. Two inference configurations were evaluated: a lightweight VLM running locally on the robot for on-device processing, and a more powerful cloud-based VLM used for remote inference. Experimental results show that both configurations effectively identified diverse anomalies, demonstrating the feasibility of combining quadruped platforms with VLM-based zero-shot perception for continuous monitoring in dynamic environments. Furthermore, a comprehensive benchmarking study was conducted across multiple state-of-the-art VLMs Gemini-2.5-Pro, GPT-5, Claude Sonnet 4.5, Gemma3-27B, Qwen3-VL-30B, and LLaVA-13B alongside human evaluators. This comparison enabled systematic assessment of model behavior and alignment with human judgment, providing deeper insights into the strengths and limitations of modern VLMs for embodied perception tasks.

INDEX TERMS Robotics, quadruped robots, vision-language models (VLMs), zero-shot anomaly detection, embodied AI, multimodal perception, autonomous patrol, ROS2.

I. INTRODUCTION

The integration of robotics across various sectors has significantly enhanced operational efficiency, particularly in environments requiring continuous monitoring and maintenance [1]. Among different robotic platforms, quadruped robots have attracted substantial attention due to their superior mobility and adaptability in complex terrains [2], [3]. Inspired by the locomotion mechanisms of four-legged animals [4], these platforms are equipped with advanced

sensors and camera systems, enabling them to navigate challenging environments and execute tasks such as inspection, surveillance, and data collection [5]. Anomalous events, defined as deviations from expected behavioral patterns, pose significant challenges in dynamic and unstructured environments [6]. Traditional anomaly detection methods predominantly rely on predefined rules or thresholds, which often fall short of capturing the complexity and diversity of real-world scenarios [7]. Recent technological advancements have fostered the adoption of Large Language Models (LLMs) for anomaly detection, leveraging their ability to comprehend and interpret complex contextual information.

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LLMs have demonstrated promising performance in identifying semantic anomalies by learning high-level relationships and patterns [8]. In parallel, rapid progress in natural language processing and vision-language learning has led to the emergence of vision-language pretraining and zero-shot inference paradigms [9]. Within this framework, Vision Language Models (VLMs) are trained on large-scale image-text datasets collected from the internet, enabling them to generalize across diverse visual recognition tasks without additional fine-tuning [10]. Through contrastive and alignment-based learning objectives, these models acquire rich visual-semantic representations, allowing zero-shot inference on previously unseen data [11], [12]. This paradigm offers a scalable alternative to task-specific supervised learning, with improved generalization capabilities [13]. The integration of quadruped robotic platforms with advanced language and vision-language models introduces a novel paradigm for intelligent anomaly detection. Quadruped robots provide high mobility and multi-modal sensing capabilities for comprehensive environmental data acquisition, while LLMs and VLMs enable advanced semantic and visual interpretation. This synergy facilitates real-time anomaly detection and enhanced situational awareness, improving operational safety and reliability in both indoor and outdoor environments.

This study investigates the use of VLMs in conjunction with quadruped robots for anomaly detection in realistic deployment scenarios. Specifically, it evaluates the ability of VLMs to interpret sensory data collected by quadruped platforms and assesses their performance in dynamic indoor and outdoor environments. The findings of this research are expected to contribute to the development of autonomous robotic systems with enhanced anomaly detection capabilities, supporting applications in industrial inspection, surveillance, environmental monitoring, and other high-risk domains.

A. RELATED WORKS

Autonomous anomaly detection in dynamic environments has been an active area of research spanning robotics, computer vision, and machine learning [14]. Traditional methods often rely on predefined rules, statistical thresholds, or classical machine learning techniques to detect deviations from normal behavior [15]. For instance, visual anomaly detection using convolutional neural networks (CNNs) has been widely studied for industrial inspection and surveillance tasks, where objects or human behaviors that deviate from training data are flagged as anomalies [16]. However, these approaches are typically limited to static camera setups or require extensive labeled datasets for effective training.

Beyond CNN-based frameworks, a growing body of work has explored advanced deep learning architectures that enhance anomaly detection performance across diverse domains. Foundational surveys highlight the benefits of deep neural models, including deep belief networks, autoencoders, and recurrent architectures, in learning hierarchical

representations for detecting abnormal patterns [17]. More recent methods introduce sophisticated generative and attention-based mechanisms. Frequency-to-spectrum mapping GANs leverage fractional Fourier-domain transformations to amplify subtle spectral deviations and reduce the likelihood of reconstructing anomalous targets [18]. Other hyperspectral approaches integrate robust PCA with autoencoder-based subspace modeling to strengthen background reconstruction and improve anomaly separability [19]. In video anomaly detection, attention-guided generators with dual discriminators employ spatial and channel attention modules to localize irregular events more effectively [20]. Furthermore, one-step detection paradigms introduce global and local self-attention mechanisms to capture universal spectral deviation relationships, enabling strong generalization to unseen data [21]. Collectively, these studies demonstrate how architectural innovations—such as spectral-domain transformations, attention-guided feature refinement, and multi-discriminator training—can substantially improve anomaly detection capability.

Recent advancements have increasingly employed mobile robots to perform anomaly detection in complex and dynamic environments [22], [23]. Mobile platforms, including wheeled and quadruped robots, offer enhanced mobility, enabling the collection of sensor data from multiple viewpoints and locations [24]. Quadruped robots, in particular, have demonstrated remarkable adaptability in traversing irregular terrains and accessing areas that are difficult for wheeled robots or stationary cameras. This capability enables continuous patrolling and environmental monitoring, which is especially important for applications such as industrial inspection, campus surveillance, office environments, and healthcare facilities [25], [26], [27], [28].

Concurrently, LLMs have been explored as tools for context-aware anomaly detection [29]. Unlike traditional visual anomaly detection methods, LLMs can analyze and interpret multimodal sensor data—including images, video, and textual metadata—providing semantically rich insights into unusual events [30]. LLM-based approaches have demonstrated the ability to detect anomalies not only by identifying statistical outliers but also by understanding the semantic context of a situation, such as unsafe human behaviors or abnormal object placements [31], [32]. Cloud-based LLMs, such as Gemini, provide high computational capacity for complex inference, while lightweight on-device LLMs enable real-time decision-making on autonomous robotic platforms [33].

The integration of quadruped robots with LLMs and VLMs represents a novel approach to autonomous anomaly detection. Prior studies have demonstrated the potential of quadruped robots for surveillance and environmental monitoring; however, these works typically rely on classical rule-based or CNN-based detection pipelines. Incorporating VLMs enables contextual reasoning about the environment, supports interpretable anomaly explanations, and allows robots to operate autonomously with increased intelligence and flexibility.

Existing anomaly detection approaches can be broadly categorized into: (i) classical rule-based and statistical methods, (ii) CNN-based visual anomaly detectors, (iii) generative and attention-driven deep learning models, and (iv) mobile robot sensing-based methods. Rule-based systems generally lack generalization capability, while CNN-based and generative models require large labeled datasets and struggle to adapt to unseen anomalies. Attention-based video models often assume static camera setups, and mobile robotic studies, although improving coverage and mobility, typically depend on conventional perception pipelines. Moreover, recent VLM-based anomaly detection studies primarily focus on static datasets or controlled laboratory conditions rather than autonomous patrol in mixed indoor-outdoor environments. None of these baselines provide a unified framework that integrates VLM reasoning, multi-environment adaptability, anomaly deduplication, and fallback mechanisms on an autonomous quadruped robot.

Despite these advancements, systematic evaluations of VLM-based anomaly detection on autonomous quadruped robots operating in both indoor and outdoor environments remain limited. This study addresses this gap by combining a quadruped robot with both lightweight on-device VLMs and cloud-based VLM inference, demonstrating the feasibility, versatility, and practical applicability of the proposed approach in real-world deployments.

B. MOTIVATION AND CONTRIBUTIONS

Continuous anomaly detection in dynamic environments is critical for ensuring safety, security, and operational efficiency in domains such as industrial facilities, healthcare environments, educational campuses, and public spaces. Conventional monitoring systems based on stationary cameras or rule-based sensors suffer from limited mobility, insufficient contextual awareness, and weak adaptability to changing conditions. In contrast, mobile robotic platforms, particularly quadruped robots, can autonomously traverse complex indoor and outdoor environments, enabling persistent and context-aware monitoring. However, existing robotic anomaly detection approaches remain largely constrained to predefined rules or low-level visual features, limiting their ability to reason about the semantic significance of observed events.

Recent advances in VLMs offer new opportunities to overcome these limitations by enabling multimodal perception and high-level semantic reasoning. When integrated with mobile robots, VLMs allow anomalies to be detected and interpreted in a context-aware manner. Nevertheless, deploying VLMs in real-world robotic systems introduces several challenges, including output variability, misclassification, and the need for robust mechanisms such as environment-aware category mapping, anomaly deduplication, and reliable fallback strategies.

Despite rapid progress in VLM-based perception, existing studies predominantly focus on static camera-based anomaly detection or evaluate VLMs in isolated and highly controlled robotic scenarios. Consequently, a clear research gap

remains in: (i) systematically integrating VLMs into mobile robotic patrol systems, (ii) enabling zero-shot anomaly detection across diverse real-world environments, and (iii) benchmarking VLM-based decisions against human-labeled ground truth. Moreover, prior approaches generally lack explicit mechanisms to ensure output consistency, mitigate noisy predictions, or adapt perception pipelines to different environment types.

In response to these limitations, this work introduces a unified framework that integrates both cloud-based and on-device VLMs with quadruped robotic platforms for zero-shot anomaly detection in diverse environments. The proposed framework incorporates mechanisms for environment-aware category mapping, anomaly deduplication, consistency verification, and flexible fallback reasoning. The effectiveness of the approach is validated through extensive real-world experiments and human-referenced evaluation.

The core contributions of this work, summarized for clarity and immediate impact, are as follows:

- A unified VLM-integrated quadruped anomaly detection framework is presented, enabling autonomous, real-time, zero-shot perception in both indoor and outdoor environments.
- The practical feasibility of deploying a VLM on a quadruped robot using an onboard GPU is demonstrated, enabling real-time, on-device inference without reliance on cloud connectivity.
- A comparative evaluation of two inference configurations—an on-device lightweight VLM and a cloud-based high-capacity VLM—is conducted under identical operational conditions.
- Comprehensive experimental validation is performed across corridors, offices, shared indoor spaces, and outdoor patrol areas, demonstrating robustness and adaptability in diverse environments.
- Multiple state-of-the-art VLMs, including Gemini-2.5-Pro, GPT-5, Claude Sonnet 4.5, Gemma3-27B, Qwen3-VL-30B, and LLaVA-13B, are benchmarked alongside human evaluators to analyze model-human alignment in embodied perception tasks.

II. METHODOLOGY

A. SYSTEM OVERVIEW OF THE ROBOT PLATFORM

The robotic platform employed in this study is the Unitree Go2 EDU+, a quadruped robot designed for versatile operation in both indoor and outdoor environments. The robot measures approximately $70 \times 31 \times 40$ cm in a standing posture and $76 \times 31 \times 20$ cm when crouching, with a total weight of approximately 15 kg, including the battery. Its mechanical structure is constructed from an aluminum alloy combined with high-strength engineering plastics, ensuring durability while maintaining a lightweight form factor. The system operates within a voltage range of 28–33.6 V and provides a peak power capacity of approximately 3000 W. It supports a nominal payload of about 8 kg, with a maximum payload capacity of up to 12 kg. The platform achieves locomotion speeds ranging from 0 to 3.7 m/s, with peak velocities

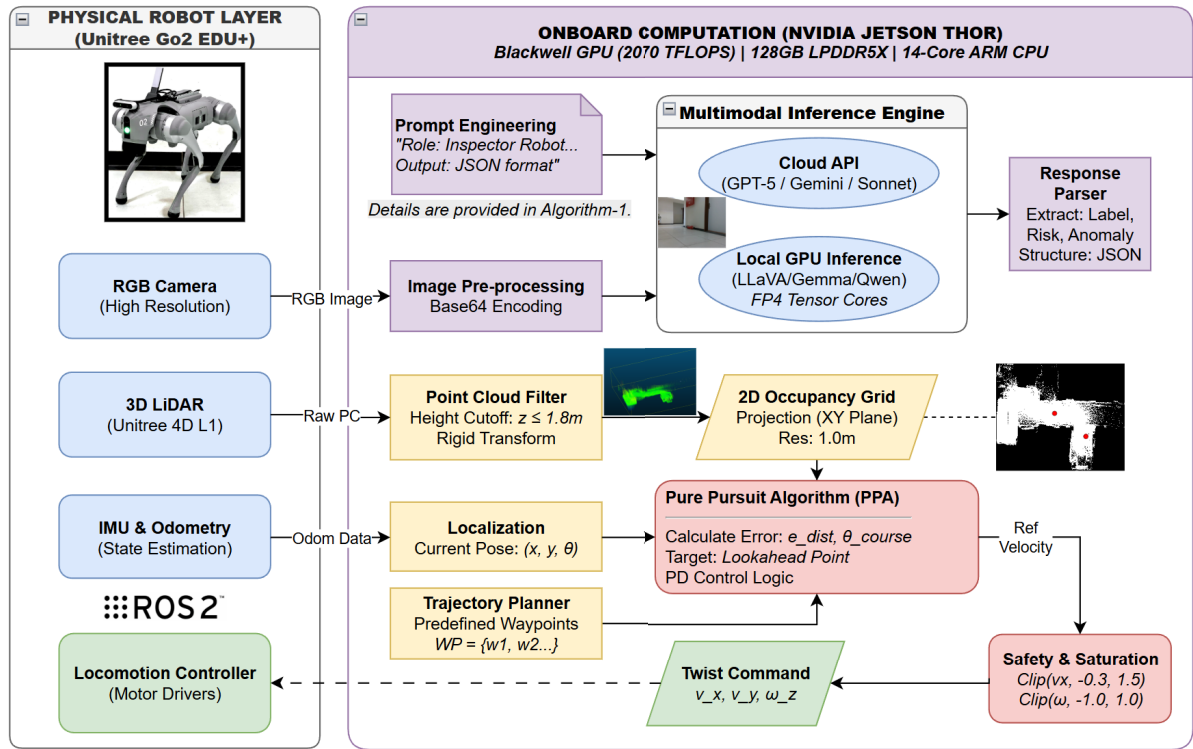


FIGURE 1. Block diagram of the proposed robotic system architecture integrating anomaly detection with navigation.

reaching approximately 5 m/s. For onboard computation, the experimental platform is equipped with an NVIDIA Jetson module serving as the embedded edge computing unit. This module combines a GPU-accelerated architecture with a multi-core Arm CPU, providing a compact and energy-efficient solution for real-time robotic perception and multimodal AI workloads. Its integrated memory subsystem supports sustained processing of high-rate visual and spatial sensor data required for autonomous patrol operations. In addition, the robotic platform is equipped with an Intel RealSense D435i RGB-D camera, which provides synchronized color and depth measurements for perception and environment understanding.

B. PROPOSED SYSTEM ARCHITECTURE

The overall architecture of the proposed robotic system for anomaly detection and autonomous navigation is illustrated in Fig. 1. The process begins with the onboard RGB camera capturing visual information from the environment, which is subsequently encoded in Base64 format for digital transmission. The encoded image data are processed by the VLM, which serves as the core decision-making unit of the system. The VLM generates a structured response that includes the identified object label (e.g., fire extinguisher), its condition (e.g., fallen on the floor), and an associated risk level (e.g., medium). In parallel, the robot acquires pose and point cloud data through a ROS-based architecture. The point cloud, represented as a set of spatial data points, is utilized to construct a two-dimensional occupancy grid

map of the surrounding environment. This map incorporates predefined waypoints to support navigation tasks. The robot's current pose, combined with the occupancy grid map, is then provided to a Pure Pursuit Algorithm (PPA) module [34], which computes the control commands required for path following and motion execution.

1) PURE PURSUIT CONTROL FOR QUADRUPED LOCOMOTION

Fig. 2 presents an example environment map and the corresponding predefined patrol trajectory. A 3D point cloud acquired from onboard sensors is projected onto the horizontal plane to generate a 2D occupancy grid map, on which waypoints are defined to guide the robot through corridors and obstacle-rich areas. Trajectory tracking is achieved using a Pure Pursuit-based controller, which computes body-level velocity commands from the robot's current pose and the target waypoint. The controller regulates forward, lateral, and angular velocities, while a dedicated low-level locomotion controller handles gait generation and stability. The overall control procedure is summarized in Algorithm 1. During patrol operation, the robot follows the predefined path at a reduced speed and pauses at each waypoint to capture images, ensuring stable and consistent visual input for VLM-based anomaly analysis. The same navigation strategy is applicable to both indoor and outdoor environments; however, outdoor experiments were conducted under manual supervision due to operational constraints.

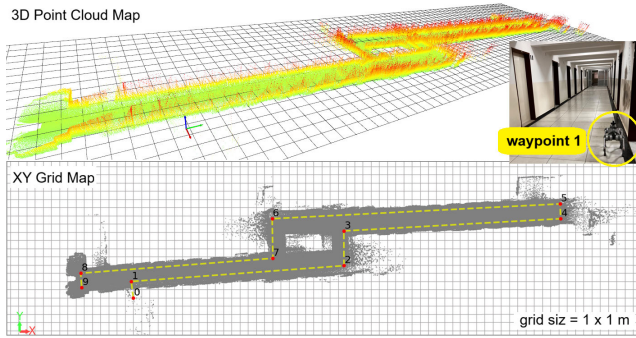


FIGURE 2. Sample floor map illustrating predefined robot trajectory. Top figure shows 3D point cloud map of environment, bottom figure presents the corresponding 2D grid map with waypoints and planned path.

Algorithm 1 Control Algorithm for Quadruped Robot

Require: Waypoint w_p , Time step dt

Ensure: Command velocities (x_{vel} , y_{vel} , ω_{vel}) and distance error e_{dist}

- 1: *Step 1: Get current pose and orientation*
- 2: $pose \leftarrow pipeline_state.qpos[0 : 3]$
- 3: $rpy \leftarrow quat_to_euler(pipeline_state.x.rot[0])$
- 4: $yaw \leftarrow rpy[2]$
- 5: *Step 2: Compute distance and heading error*
- 6: $e_{dist} \leftarrow \|pose - wp\|$
- 7: $course \leftarrow \arctan 2(wp_y - pose_y, wp_x - pose_x)$
- 8: $\theta_c \leftarrow (course - yaw + \pi) \bmod (2\pi) - \pi$
- 9: *Step 3: Calculate reference velocities*
- 10: $v_{ref} \leftarrow bias + k_{px}e_{dist} + k_{dx}\frac{\Delta e_{dist}}{dt} - k_r|\theta_c|$
- 11: $y_{ref} \leftarrow k_{py}\theta_c$
- 12: $\omega_{ref} \leftarrow k_{pw}\theta_c + k_{dw}\frac{\Delta \theta_c}{dt}$
- 13: *Step 4: Apply saturation limits*
- 14: $x_{vel} \leftarrow clip(v_{ref}, -0.3, 1.5)$
- 15: $y_{vel} \leftarrow clip(y_{ref}, -0.8, 0.8)$
- 16: $\omega_{vel} \leftarrow clip(\omega_{ref}, -1.0, 1.0)$
- 17: **return** (x_{vel} , y_{vel} , ω_{vel} , e_{dist})

It should be emphasized that navigation is included solely to support autonomous patrol and consistent data acquisition; the primary contribution of this study lies in the perception and semantic reasoning pipeline enabled by VLMs.

2) VLM-DRIVEN ANOMALY DETECTION

The core perception and decision-making capability of the proposed system is realized through VLMs, which enable high-level semantic interpretation of visual observations without task-specific training. Unlike traditional vision-based anomaly detection methods that rely on predefined classes or handcrafted features, the VLM-based approach allows anomalies to be identified and described in a zero-shot and context-aware manner. During autonomous patrol, the quadruped robot acquires RGB observations at predefined spatial locations, and each image is analyzed independently by the VLM to assess object presence, condition, and associated risk. To enable spatial reasoning and traceability,

Algorithm 2 Perception-Level VLM-Based Anomaly Detection

Require: RGB image from robot camera

Ensure: List of detected anomalies with risk levels

- 1: Acquire and Base64-encode RGB image
- 2: Detect objects using VLM
- 3: **Prompt used:**

"You are an inspector robot. The image is taken from a camera mounted on a quadruped robot. Your task is to determine potential anomalies such as fallen objects, spills, puddles, broken glass, loose cables, missing fire extinguisher, obstructed emergency exits. For outdoor environments, also consider anomalies such as wrongly parked vehicles, broken streetlights, fallen barriers, chemical leaks, missing safety helmets. For each anomaly detected, respond in the following format: **environment:** <indoor/outdoor> | **label:** <object label> | **anomaly:** <short description> | **risk:** <low/medium/high>. Keep the anomaly response as short as possible (max 8 words each). If no anomaly is detected, respond: 'No anomaly detected.'"

- 4: **return** final anomaly list

detected anomalies are associated with the robot's estimated pose obtained via LiDAR-based localization within a three-dimensional environment.

At each predefined waypoint, the robot executes a structured inspection procedure. Upon reaching the waypoint, locomotion is halted and the robot stabilizes its posture. The robot then adjusts its head orientation to an upward-facing configuration to maximize the observable field of view, particularly for elevated or partially occluded anomalies. After posture stabilization, a single RGB frame is captured and forwarded to the VLM for semantic anomaly analysis. This inspection routine is repeated consistently at all waypoints, ensuring uniform observation conditions across scenarios.

The VLM-based anomaly detection algorithm was intentionally designed as a zero-shot, prompt-driven pipeline to avoid task-specific fine-tuning and enable rapid deployment across diverse environments. The fixed output schema enforces structural consistency across model responses, facilitating quantitative comparison and automated evaluation. By decoupling perception from navigation, the proposed design ensures that failures or uncertainties in semantic reasoning do not propagate into motion control.

The pseudo-code of the algorithm is provided in Algorithm 2. In this study, a Python-based pipeline was developed to automate the evaluation of anomaly detection performance using VLMs. A structured textual prompt guides the VLM in identifying anomalies from images captured by a quadruped robot, specifying both indoor and outdoor scenarios and defining a fixed response format consisting of environment, object label, anomaly description, risk level, and recommended action. Each image is read from a predefined directory, encoded in Base64 format, and included in the API request together with the prompt to enable multimodal reasoning. To ensure robust communication, an exponential backoff retry strategy is employed in the event of transient HTTP errors, with up to five attempts per image. The responses are parsed from the JSON output, with newline characters removed to produce compact, single-line entries,

and any empty or malformed responses are logged for further inspection. This pipeline enables the systematic collection of VLM outputs for subsequent performance analysis.

III. RESULTS

A. OVERVIEW OF EVALUATION SCENARIOS

Six state-of-the-art VLMs, namely Gemini-2.5-Pro, GPT-5, Claude Sonnet 4.5, Gemma3-27B, Qwen3-VL-30B, and LLaVA-13B, were evaluated for their ability to detect anomalies in images collected by a quadruped robot. Human assessments served as the reference for performance comparison. The evaluation encompassed 20 scenarios across both indoor and outdoor environments, capturing a wide range of anomaly types, including misplaced objects, environmental hazards, and structural irregularities. Several anomalies were manually introduced to ensure controlled diversity across test conditions. Detailed model responses for all scenarios are provided in the Appendix (Table 4 and Table 5) for completeness.

Fig. 3 presents a representative subset of the evaluated scenes, each labeled with a scenario identifier (s_1 – s_{20}).

B. DATASET AND EVALUATION PROTOCOL

The experimental dataset consists of 20 distinct scenes, each represented by a single RGB image captured by the onboard camera of the quadruped robot. For each scene, only one image was acquired, and no repeated captures were performed. All images were collected under naturally occurring lighting conditions without any artificial illumination or environmental manipulation. No preprocessing or data augmentation techniques were applied to the collected images. Specifically, the raw RGB images were directly used as input to the VLMs without resizing, cropping, normalization, or color adjustment. Each image was encoded in Base64 format solely for transmission purposes and was otherwise unmodified. All evaluated VLMs were tested individually on the exact same set of images to ensure a fair and consistent comparison across models. Each image was presented to a given model only once. This design choice was motivated by the well-known tendency of large language models to reproduce previous responses when queried multiple times with the same input. Consequently, the primary objective of the evaluation was to assess how each VLM interprets and reasons about a previously unseen image during its first exposure. For human baseline validation, 10 independent human evaluators were recruited. Each participant reviewed the same set of images and annotated the presence or absence of anomalies, as well as their perceived risk levels. The aggregated human annotations were used as ground truth for quantitative evaluation. Model performance was assessed using accuracy as the primary metric. All anomaly classes and scenarios were assigned equal weight in the accuracy computation, ensuring that no particular anomaly type dominated the evaluation results. This uniform weighting strategy enables a balanced and transparent comparison of anomaly detection performance across different VLMs.

C. QUANTITATIVE COMPARISON WITH HUMAN BASELINE

A comprehensive quantitative comparison across all 20 scenarios and four evaluation criteria environment understanding (en), label consistency (lb), anomaly recognition (an), and risk reasoning (rs) is summarized in Table 1. Human annotations serve as the gold-standard reference. Gemini-2.5-Pro and GPT-5 demonstrate the highest overall alignment with human judgment, exhibiting strong semantic consistency and robust performance across both indoor and outdoor environments. Claude Sonnet 4.5 also performs reliably but occasionally shows mild over-interpretation, manifested as pluralized or overly detailed descriptions. In contrast, Gemma3-27B, Qwen3-VL-30B, and LLaVA-13B exhibit reduced consistency, particularly in visually complex or low-contrast scenes. Across indoor scenarios involving spills, fallen objects, and physical obstructions, Gemini-2.5-Pro, GPT-5, and Claude Sonnet 4.5 achieve the highest contextual accuracy. Gemini-2.5-Pro further demonstrates strong generalization in outdoor settings, successfully identifying anomalies such as overgrown vegetation and obstructed drainage paths. While Claude Sonnet 4.5 benefits from strong language fluency, this occasionally comes at the cost of weaker visual grounding. Gemma3-27B provides stable but comparatively less descriptive outputs across most scenarios.

D. STATISTICAL ANALYSIS OF VLM PERFORMANCE

Although accuracy provides an intuitive measure of overall correctness, it does not fully capture potential model biases or differences in error severity across evaluation criteria. Metrics such as precision, recall, or F1-score were not adopted because the task involves multi-attribute reasoning environment, label, anomaly, and risk where each attribute is evaluated as an independent binary outcome rather than a class-based prediction. Under this formulation, class-oriented metrics offer limited interpretability. Accordingly, accuracy was selected as the most consistent and interpretable metric across heterogeneous evaluation dimensions. Human annotations were treated as the reference ground truth for all quantitative analyses.

The performance of the VLMs was evaluated quantitatively using accuracy, which measures the proportion of correct predictions relative to the total number of predictions. The general formulation of accuracy is given in Eq. 1.

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}} \times 100 \quad (1)$$

The total accuracy for each model was computed by averaging the accuracy across all evaluation criteria. To ensure a fair and reproducible comparison, all VLMs were evaluated by the authors using the same images, identical prompts, and a unified accuracy-based evaluation protocol. No external benchmarks or previously reported results were used.

Table 2 presents the accuracy percentages for each model across all criteria. Gemini-2.5-Pro and GPT-5 achieved the highest overall agreement with the human baseline, indicating robust performance across most scenarios. Claude Sonnet 4.5 exhibited occasional hallucinations, particularly



FIGURE 3. Examples of anomalies in indoor and outdoor environments used to evaluate the performance of VLMs. See the Appendices for details.

TABLE 1. Comparison of VLMs on 20 scenarios with detailed anomaly detection outputs.

Scenario	Gemini-2.5-Pro				GPT-5				Claude Sonnet 4.5				Gemma3-27B				Qwen3-VL-30B				LLaVA-13B				Human				
	en	lb	an	rs	en	lb	an	rs	en	lb	an	rs	en	lb	an	rs	en	lb	an	rs	en	lb	an	rs	en	lb	an	rs	
S1	+	+	+	+	-	+	+	+	-	p	+	+	+	+	+	+	+	+	+	+	-	+	-	+	+	+	+	+	+
S2	+	+	+	+	+	+	+	+	+	p	+	+	+	+	+	+	+	+	+	+	-	-	-	+	+	+	+	+	+
S3	+	+	+	-	+	+	+	+	+	+	+	+	-	+	-	+	+	+	+	+	+	+	+	-	+	+	+	+	+
S4	+	+	+	+	-	-	-	-	+	+	+	+	+	+	-	-	+	+	-	-	+	-	-	-	+	+	+	+	+
S5	+	-	+	+	-	-	-	-	+	-	-	-	-	+	-	-	-	+	-	-	+	-	-	-	+	+	+	+	+
S6	+	+	+	+	+	+	+	-	+	+	+	+	-	+	+	+	+	+	+	+	+	-	+	-	+	+	+	+	+
S7	+	+	+	-	+	+	+	-	+	+	+	-	-	+	+	+	+	-	+	+	+	+	+	-	+	+	+	+	+
S8	+	+	+	-	+	-	+	-	+	+	+	+	+	+	+	+	-	+	+	+	-	+	-	+	-	+	+	+	+
S9	+	+	+	-	+	+	+	-	+	+	+	+	-	+	+	+	-	+	+	+	-	+	-	+	+	+	+	+	+
S10	-	-	-	-	na	na	na	na	-	-	-	-	-	-	-	-	-	na	na	na	na	-	-	-	-	na	na	na	na
S11	+	+	-	-	+	+	+	-	-	-	-	-	-	-	-	-	-	-	-	-	+	-	-	+	+	+	+	+	+
S12	+	+	+	+	+	+	+	+	+	-	-	+	+	+	+	-	+	+	-	+	+	+	-	+	+	+	+	+	+
S13	+	+	+	-	+	+	+	-	+	+	+	+	+	+	+	-	+	+	+	+	+	-	-	-	+	+	+	+	+
S14	+	+	+	+	+	+	+	-	+	+	+	+	-	+	+	+	+	+	+	+	+	-	-	+	+	+	+	+	+
S15	+	+	-	-	+	+	+	-	+	+	+	+	+	+	+	+	+	-	-	-	-	+	-	-	+	+	+	+	+
S16	+	+	+	-	+	+	+	+	+	+	+	+	+	+	+	+	-	+	+	+	-	+	+	+	-	+	+	+	+
S17	+	+	+	-	+	+	+	+	+	-	-	-	-	+	+	+	-	+	+	+	-	+	+	-	+	+	+	+	+
S18	+	+	+	-	+	+	+	-	+	+	+	+	+	+	+	+	-	+	+	-	-	+	-	-	+	+	+	+	+
S19	+	+	+	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	+	+	+	+	+
S20	+	+	+	-	+	+	+	-	+	+	-	-	-	+	-	-	-	-	-	-	+	-	-	-	+	+	+	+	+

Abbreviations: environment (en), label (lb), anomaly (an), risk (rs), and no anomaly detected (na). true (+), and false (-), plural hallucination (p).

in label consistency and anomaly recognition. Gemma3-27B, Qwen3-VL-30B, and LLaVA-13B demonstrated variable performance depending on the scenario. Certain scenarios, such as s_{10} and s_{19} , were consistently challenging for most models.

E. INFERENCE TIME AND DEPLOYMENT ANALYSIS

The results summarized in Table 3 reveal substantial latency differences across VLM architectures, particularly when deployed on embedded hardware. Although Gemma3-27B

and Qwen3-VL-30B have comparable parameter sizes, their inference times vary markedly on Jetson Thor. This discrepancy arises from architectural characteristics rather than model size alone. Qwen3-VL-30B incorporates a high-capacity ViT-L vision encoder together with extensive cross-modal attention layers, generating a large number of visual tokens and significantly increasing memory traffic during inference. On platforms with limited memory bandwidth, such as Jetson Thor, this design introduces a memory-bound bottleneck, resulting in pronounced latency spikes (e.g., 20 s). In contrast, Gemma3-27B and LLaVA-13B employ lighter

TABLE 2. Accuracy (%) of each VLM and human baseline across four evaluation criteria.

Model	en	lb	an	rs	Total Accuracy
Gemini-2.5-Pro	95	85	85	35	75.0
GPT-5	80	80	85	35	70.0
Claude Sonnet 4.5	80	55	60	40	58.7
Gemma3-27B	85	60	55	30	57.5
Qwen3-VL-30B	80	65	55	45	61.2
LLaVA-13B	75	30	25	30	40.0
Human (baseline)	100	100	100	100	100

Abbreviations: environment (en), label (lb), anomaly (an), risk (rs)

TABLE 3. Comparison of anomaly detection inference time across cloud-based and local VLM configurations.

Model	Cloud (API)	RTX 5090	Jetson Thor
Gemini-2.5-Pro	3.7415	—	—
GPT-5	2.4490	—	—
Claude Sonnet 4.5	3.2653	—	—
Gemma3-27B	—	3.0830	5.6545
Qwen3-VL-30B	—	4.7100	11.2785
LLaVA-13B	—	2.6875	3.7005

All inference times are reported in seconds (s).

visual encoders, enabling faster and more stable inference under identical hardware constraints. These results indicate that model architecture and visual tokenization strategy play a more critical role in embedded deployment feasibility than parameter count alone.

F. FAILURE CASE ANALYSIS AND DISCUSSION

In particular, Gemini-2.5-Pro and GPT-5 achieved the highest anomaly identification accuracy, reaching 85% on this criterion. In contrast, all models exhibited substantially lower accuracy in risk reasoning, largely due to the absence of task-specific prior guidance regarding risk levels during evaluation. Examination of scenarios s_8 and s_{13} illustrates this limitation: standing water on the floor creates a slippery surface and should be classified as a high-risk condition, yet only Claude Sonnet 4.5 correctly inferred this level of danger. Similarly, in scenario s_{16} , the exposed electrical panel and dangling live cable represent a clear electrical hazard, but only GPT-5 identified the situation as high risk.

The consistently low performance in risk reasoning across all models indicates that current VLMs have limited capability to map visual cues to implicit safety semantics. While anomaly detection can primarily rely on observable visual features, risk assessment requires an understanding of regulatory norms, domain-specific thresholds, and causal relationships. In the absence of such priors, general-purpose VLMs tend to generate under-specified or overly generic safety judgments.

Overall, the findings suggest that standard VLMs are not yet capable of performing reliable and consistent risk analysis without domain-specific grounding. This limitation may be alleviated by incorporating occupational safety guidelines or similar regulatory knowledge into the model context or fine-tuning pipeline prior to deployment.

The analyzed scenarios further reveal common visual failure modes. Standing water in s_8 and s_{13} exhibits low contrast and lacks well-defined object boundaries,

making it difficult for vision encoders to recognize it as a hazardous condition. Likewise, the exposed electrical panel in s_{16} requires fine-grained structural interpretation and domain-specific safety awareness to be classified as high risk. Such visually subtle and context-dependent hazards expose fundamental limitations of existing VLMs, which tend to prioritize salient objects over implicit environmental risks.

A cross-scenario analysis shows consistent error patterns across different models. Most false negatives occurred when anomalies were visually subtle, partially occluded, or blended into the background. In addition, reflective or transparent surfaces were occasionally misinterpreted as normal environmental elements. These trends indicate that even high-performing VLMs remain strongly dependent on salient visual cues and lack robust mechanisms for identifying low-saliency or context-dependent hazards.

IV. CONCLUSION

This study presented an integrated framework that combines state-of-the-art VLMs with a quadruped robotic platform to enable real-time anomaly detection across diverse indoor and outdoor environments. By leveraging RGB imagery alongside LiDAR-based localization, the proposed system achieves multimodal, context-aware perception and semantic reasoning in an embodied robotic setting. A comparative evaluation across multiple VLMs demonstrated that Gemini-2.5-Pro, GPT-5, and Claude Sonnet 4.5 achieved the highest overall alignment with human judgment, particularly in scenarios involving multiple simultaneous anomalies. Among the open-source models, Gemma3-27B exhibited a favorable balance between computational efficiency and detection performance, highlighting its suitability for embedded robotic deployment. Overall, the results confirm the effectiveness of integrating advanced VLM-based reasoning with mobile robotic perception to enhance situational awareness and continuous monitoring in safety-critical environments.

Future work will focus on improving system robustness and autonomy. The integration of additional sensing modalities, such as thermal and depth imaging, is expected to enhance detection reliability under challenging environmental conditions. Another promising direction involves the development of lightweight, on-device VLM architectures, which would reduce reliance on cloud-based inference and enable fully autonomous operation. With ongoing advances in high-performance embedded computing for robotics, deploying VLMs directly on mobile platforms is becoming increasingly feasible.

APPENDIX ANOMALY DETECTION RESULT TABLES

See Tables 4 and 5.

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TABLE 4. Anomaly detection results on 1-10 images (see Fig. 3) for different VLM models.

Scene	Model	environment	label	anomaly	risk
S1	Gemini-2.5-Pro	outdoor	trash bag	Large bag obstructing a doorway.	medium
	GPT-5	indoor	trash bag	bag blocking doorway	medium
	Sonnet 4.5	indoor	trash bags	bags blocking doorway/exit	medium
	Gemma-27B	outdoor	trash bag	obstructed pathway	low
	Qwen3-VL-30B	outdoor	black bag	fallen object	medium
	LLaVA-13B	indoor	garbage bag	full garbage bag	high
S2	Human	outdoor	garbage bag	A trash bag is blocking the doorway.	medium
	Gemini-2.5-Pro	outdoor	trash bag	Obstructing entrance, creating a trip hazard.	medium
	GPT-5	outdoor	trash bag	bag placed on entrance steps	medium
	Sonnet 4.5	outdoor	trash bags	bags on stairs near entrance	medium
	Gemma-27B	outdoor	debris	object obstructing stairs	medium
	Qwen3-VL-30B	outdoor	black bag	obstructed emergency exit	medium
S3	LLaVA-13B	indoor	steps	broken	medium
	Human	outdoor	garbage bag	A trash bag is blocking the doorway.	medium
	Gemini-2.5-Pro	indoor	trash bag	Object on floor, potential trip hazard.	low
	GPT-5	indoor	trash bag	garbage bag near exit	medium
	Sonnet 4.5	indoor	trash bag	bag left in corridor area	low
	Gemma-27B	indoor	trash	trash on floor	low
S4	Qwen3-VL-30B	indoor	bag	Fallen bag on floor	low
	LLaVA-13B	indoor	trash bag	black trash bag on floor	low
	Human	indoor	garbage bag	Trash bags should not be kept indoors.	medium
	Gemini-2.5-Pro	outdoor	debris	Litter and leaves scattered on the ground.	low
	GPT-5	-	-	No anomaly detected.	-
	Sonnet 4.5	outdoor	fallen leaves	heavy leaf accumulation on pavement	low
S5	Gemma-27B	outdoor	vehicle	wrongly parked motorcycle	low
	Qwen3-VL-30B	outdoor	spill	white spill on ground	low
	LLaVA-13B	-	-	No anomaly detected.	-
	Human	outdoor	leaf pile	There should not be a pile of leaves in the parking area.	low
	Gemini-2.5-Pro	outdoor	ground	Excessive leaves on unpaved ground	low
	GPT-5	-	-	No anomaly detected.	-
S6	Sonnet 4.5	outdoor	vehicle parking	car parked on sidewalk area	medium
	Gemma-27B	outdoor	motorcycle	parked on pedestrian walkway	medium
	Qwen3-VL-30B	outdoor	vehicle	wrongly parked	medium
	LLaVA-13B	outdoor	parked vehicles	wrongly parked vehicles	medium
	Human	outdoor	leaf pile	There should not be a pile of leaves in the parking area.	low
	Gemini-2.5-Pro	outdoor	Debris	Large accumulation of trash and cardboard	medium
S7	GPT-5	outdoor	trash	litter scattered on ground	low
	Sonnet 4.5	outdoor	scattered trash	plastic bags and debris on ground	low
	Gemma-27B	outdoor	debris	scattered trash	low
	Qwen3-VL-30B	outdoor	waste	pile of cardboard/trash	medium
	LLaVA-13B	outdoor	building	trash on ground	high
	Human	outdoor	waste	The waste materials are not stored properly.	medium
S8	Gemini-2.5-Pro	indoor	trash can	Fallen object	low
	GPT-5	indoor	trash can	trash can overturned	low
	Sonnet 4.5	indoor	trash bin	bin tipped over in corridor	low
	Gemma-27B	indoor	bucket	fallen object	low
	Qwen3-VL-30B	indoor	trash bin	fallen trash bin	medium
	LLaVA-13B	indoor	trash can	tipped over	high
S9	Human	indoor	trash bin	The trash bin has fallen over.	medium
	Gemini-2.5-Pro	indoor	puddle	Liquid spill on the floor	medium
	GPT-5	indoor	floor	liquid spill on floor	medium
	Sonnet 4.5	indoor	water spill	large puddle on floor	high
	Gemma-27B	indoor	puddle	water spill on floor	medium
	Qwen3-VL-30B	indoor	puddle	water spill on floor	medium
S10	LLaVA-13B	indoor	floor	puddle of water	medium
	Human	indoor	puddle	There is a puddle of water on the floor.	high
	Gemini-2.5-Pro	indoor	bottle	Fallen object on the floor.	low
	GPT-5	indoor	plastic bottle	bottle fallen on floor	low
	Sonnet 4.5	indoor	plastic bottles	bottles left on floor	low
	Gemma-27B	indoor	bottle	fallen object	low
S11	Qwen3-VL-30B	indoor	bottle	fallen bottle	low
	LLaVA-13B	indoor	paper	spill on floor	medium
	Human	indoor	water bottle	A plastic water bottle has fallen on the floor.	medium
	Gemini-2.5-Pro	indoor	cables	Loose and messy cables on table.	low
	GPT-5	-	-	No anomaly detected.	-
	Sonnet 4.5	indoor	open cabinet	cluttered storage with items visible	low
S12	Gemma-27B	indoor	objects	items cluttered on shelf	low
	Qwen3-VL-30B	-	-	No anomaly detected.	-
	LLaVA-13B	indoor	table	bottle on top of paper	low
	Human	-	-	No anomaly detected.	-
	Gemini-2.5-Pro	indoor	-	-	-
	GPT-5	-	-	-	-

TABLE 5. Anomaly detection results on 11-20 images (see Fig. 3) for different VLM models.

Scene	Model	environment	label	anomaly	risk
S11	Gemini-2.5-Pro	indoor	paper/cardboard	Large paper sheet leaning against the wall.	low
	GPT-5	indoor	open electrical panel	cabinet door left open	medium
	Sonnet 4.5	-	-	No anomaly detected.	-
	Gemma-27B	-	-	No anomaly detected.	-
	Qwen3-VL-30B	-	-	No anomaly detected.	-
	LLaVA-13B	indoor	doorway	blocked	medium
S12	Human	indoor	electrical panel	The electrical panel is open.	high
	Gemini-2.5-Pro	indoor	spill	Fallen object and spilled soil on floor.	low
	GPT-5	indoor	plant pot	fallen green pot on floor	low
	Sonnet 4.5	indoor	floor clutter	items and planters on floor	low
	Gemma-27B	indoor	plant	potted plants obstructing walkway	low
	Qwen3-VL-30B	indoor	watering can	fallen watering can	low
S13	LLaVA-13B	indoor	potted plant	plant has dried up	high
	Human	indoor	flower pot	The flower pot has fallen over.	low
	Gemini-2.5-Pro	indoor	puddle	Large liquid spill on the floor.	medium
	GPT-5	indoor	water puddle	water spilled on floor	medium
	Sonnet 4.5	indoor	water on floor	significant water pooling throughout kitchen	high
	Gemma-27B	indoor	bottle	fallen bottle	low
S14	Qwen3-VL-30B	indoor	puddle	wet floor patch	high
	LLaVA-13B	indoor	floor	wetness	medium
	Human	indoor	water bottle	A water bottle tipped over and created a puddle.	high
	Gemini-2.5-Pro	indoor	chair	Fallen object creating a trip hazard.	medium
	GPT-5	indoor	chair	fallen chair on floor	low
	Sonnet 4.5	indoor	fallen chair	chair tipped over on floor	low
S15	Gemma-27B	indoor	chair	chair fallen over	medium
	Qwen3-VL-30B	indoor	chair	fallen chair	medium
	LLaVA-13B	indoor	floor	clutter	medium
	Human	indoor	chair	The chair has fallen over.	medium
	Gemini-2.5-Pro	indoor	broom	Fallen object creating a trip hazard.	low
	GPT-5	indoor	broom	broom leaning, potential trip hazard	low
S16	Sonnet 4.5	indoor	cleaning equipment	mop left in middle of corridor	medium
	Gemma-27B	indoor	mop	mop obstructing walkway	medium
	Qwen3-VL-30B	-	-	No anomaly detected.	-
	LLaVA-13B	indoor	floor lamp	fallen	high
	Human	indoor	broom	The broom is blocking the way.	medium
	Gemini-2.5-Pro	indoor	electrical panel	Panel door is open, exposing wiring.	medium
S17	GPT-5	indoor	electrical panel	open panel with loose cable	high
	Sonnet 4.5	indoor	electrical panel	panel door left open	high
	Gemma-27B	indoor	cable	loose cable hanging down	medium
	Qwen3-VL-30B	indoor	cable	loose cable	medium
	LLaVA-13B	indoor	circuit box	open circuit box on wall	low
	Human	indoor	electrical panel; cable	A loose cable is hanging from an open electrical panel.	high
S18	Gemini-2.5-Pro	indoor	fire extinguisher	Fire extinguisher fallen on the floor.	medium
	GPT-5	indoor	fire extinguisher	extinguisher lying on floor	high
	Sonnet 4.5	indoor	toy car	object obstructing hallway path	low
	Gemma-27B	indoor	fire extinguisher	extinguisher on floor	medium
	Qwen3-VL-30B	indoor	fire extinguisher	fallen fire extinguisher	medium
	LLaVA-13B	indoor	fire extinguisher	missing	high
S19	Human	indoor	fire extinguisher	The fire extinguisher has fallen over.	high
	Gemini-2.5-Pro	outdoor	drain	partially obstructed by overgrown vegetation and debris	medium
	GPT-5	outdoor	drain grate	vegetation and debris obstructing grate	medium
	Sonnet 4.5	outdoor	uncovered drainage grate	open drain with debris accumulation	high
	Gemma-27B	outdoor	debris	scattered debris around drain	low
	Qwen3-VL-30B	outdoor	cigarette butt	fallen litter	low
S20	LLaVA-13B	outdoor	street light	broken streetlight	low
	Human	outdoor	drain gate	The drain grate is obstructed by weeds	medium
	Gemini-2.5-Pro	outdoor	vegetation	overgrown vegetation and weeds on pavement	low
	GPT-5	-	-	No anomaly detected.	-
	Sonnet 4.5	-	-	No anomaly detected.	-
	Gemma-27B	-	-	No anomaly detected.	-
S21	Qwen3-VL-30B	-	-	No anomaly detected.	-
	LLaVA-13B	-	-	—— long answer	-
	Human	outdoor	weeds	Weeds have spread into the parking area.	medium
	Gemini-2.5-Pro	outdoor	drainage grate	Grate is overgrown with weeds and grass.	low
	GPT-5	outdoor	drain grate	vegetation growing in grate	low
	Sonnet 4.5	outdoor	fallen barrier/fence	collapsed metal fence section on ground	medium
S22	Gemma-27B	outdoor	broken glass	shards on ground	medium
	Qwen3-VL-30B	-	-	No anomaly detected.	-
	LLaVA-13B	outdoor	sidewalk	weeds growing in cracks	medium
	Human	outdoor	drain gate	The drain grate is obstructed by weeds	high

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